Table 1: Onset detectors over the same 140 gated runs, each carried through the same inversion. "CV" is the within-configuration coefficient of variation of the onset moment; "offset RMS" is the delivered quantity against the load cell. $\Delta|\mathrm{err}|$ is the mean paired change in $|\mathrm{error}|$ against the proposed method, with an exact sign test over the ten configurations.

method	CV mean [%]	CV worst [%]	offset RMS [mm]	mean [mm]	$\Delta\|\mathrm{err}\|$ [mm]	p
COSH (proposed)	2.9	5.8	1.65	−0.93	—	—
COSH, CAD constants	4.4	12.3	1.64	−0.86	+0.00	0.75
COSH, wide box	2.9	5.7	1.64	−0.95	−0.02	0.75
per-run NLS	6.7	19.8	1.72	−1.07	+0.02	0.34
CPD, Gaussian cost	4.2	8.6	1.67	−0.91	−0.03	0.75
CPD, RBF kernel	9.3	27.4	1.65	−0.46	−0.09	1.00
CUSUM, mean change	7.7	24.0	1.82	−0.58	+0.29	0.75

VIII. Results

Excitation campaign and gating

Five centre-of-mass configurations were produced by relocating ballast within the planar offset box, giving ten independent case/axis configurations once roll and pitch are counted separately. Each was excited in both tip directions at commanded ramp rates of 0.10, 0.30, 0.65 and 1.20 N m/s. [**TODO:** state the number of runs per configuration per direction and the total attempted]

Runs enter the identification only through the gates of Sec. VI-B: a commanded-versus-realised rate error below 3%, a linearity residual below 30 mN m RMSE about the best-fit line, and a minimum post-onset sample count. 140 runs passed and are the basis of everything below. The gates are deliberately blunt — they reject ramps that were stepped or aborted rather than tuning the fit — and no run was removed on the basis of its identified value.

Onset repeatability and the choice of detector

Within a configuration and direction, the identified onset moment repeats to a coefficient of variation of 2.9% on average and 5.8% at worst. That figure is the primary measure of the method's precision, and it is what the deviation bound of Sec. VI-E predicts should be achievable.

Because a two-segment fit could in principle be finding structure that a simpler detector would find too, six alternatives were run over the identical gated set and carried through the identical inversion (Table 1). Two are variants of the proposed estimator — one using CAD rather than identified constants, one widening the search box — and four are model-agnostic: a per-run nonlinear least squares, two change-point detectors, and a CUSUM.

Two things follow, and they point in opposite directions. The proposed estimator is the most repeatable by a wide margin — 2.9% against 4.2 to 9.3% — which matters for a pre-flight check that must return the same answer twice. But every method delivers an offset RMS between 1.64 and 1.82 mm, no paired comparison approaches significance ($p \geq 0.34$), and the delivered offsets agree to 1.09 mm in the median across all seven. The onset is therefore *not* the limiting step: the accuracy of the deliverable is set elsewhere, which Sec. VIII-D takes up.

Identified offset against the load cell

The deliverable is the pivot-free average (pivot-free avg), inverted to a centre-of-mass offset and compared against an independent load-cell measurement of the same quantity (Table 2).

Over the ten configurations the RMS error is 1.64 mm, with a bootstrap 95% confidence interval of 0.92 to 2.24 mm (20k resamples), and the worst configuration is 3.34 mm. Against offsets that span ±15 mm this is a relative accuracy of roughly one part in ten of the range identified.

The mean error is −0.91 mm with a confidence interval of −1.94 to +0.11 mm; seven of ten errors are negative, which an exact sign test does not establish ($p = 0.34$). We therefore report the negative mean as a point estimate and not as a demonstrated bias — ten configurations do not settle it, and the interval includes zero.

Table 2: Identified centre-of-mass offset against the load-cell truth, over the ten configurations. "repeat sd" is the standard deviation over run pairs within the configuration, and "err/sd" expresses the error in units of that scatter.

case	comp.	ident. [mm]	truth [mm]	error [mm]	repeat sd [mm]	err /sd
case_01	y_{off}	-1.60	-2.90	$+1.30$	1.09	1.2
case_01	x_{off}	-11.18	-11.45	$+0.27$	0.47	0.6
case_02	y_{off}	-15.15	-14.29	-0.86	0.40	2.1
case_02	x_{off}	-10.57	-9.90	-0.67	0.44	1.5
case_03	y_{off}	-8.60	-5.26	-3.34	0.51	6.5
case_03	x_{off}	0.72	3.14	-2.42	0.44	5.4
case_04	y_{off}	5.24	6.67	-1.43	0.74	1.9
case_04	x_{off}	0.22	2.40	-2.18	0.69	3.2
case_05	y_{off}	10.79	10.91	-0.12	0.82	0.2
case_05	x_{off}	-10.57	-10.89	$+0.32$	0.77	0.4

Two internal consistency checks accompany the comparison, neither of which uses the load cell. Eq. (offset eq) estimates the same offset twice, once from each tip direction with that direction's own contact arm, and (weight eq) returns the vehicle weight from the onset moments. Without a ground-effect term the two directional estimates disagree by 12.6 mm in the median and (weight eq) returns a weight 5.2% below mg in all ten configurations; restoring the interference channels brings the spread to 3.6 mm and the weight to 1.1% below. The identified offset itself is reported without a ground-effect model, to match the quantity that is flown; Sec. VI-B bounds what that omission costs at about half a millimetre either way.

What limits the accuracy

The errors of Table 2 are not run scatter. Within a configuration the run-pair standard deviation is 0.60 mm in the median and 1.09 mm at worst, whereas the standard deviation of the error *between* configurations is 1.43 mm — a ratio of 2.4. Read against each configuration's own scatter, three of the ten errors exceed three standard deviations, the worst reaching 6.5. The error is therefore systematic per configuration, and repeating a configuration cannot reduce it. This is the central statistical statement of the paper: the reported accuracy is limited by the ground-contact geometry, not by the number of excitation runs.

The estimator's own contribution is smaller by orders of magnitude. Propagating the measured out-of-span forcing through the exact kernel (Sec. VI-E) gives a residual of 2.75 mN m in the median (12.28 at worst) on the gravity channel and 0.53 (3.20) on the ground-effect channel, a rate deviation of 0.24% of the nominal in the median, and a threshold error of 0.04 mN m in the median and 0.15 at worst — 0.001 to 0.004 mm of offset. The a priori guarantee over the whole admissible operating box, which assumes nothing about the runs beyond the tilt cap and the ramp-rate range, is 0.174 mm for roll and 0.140 mm for pitch. Either figure is an order of magnitude below the 1.64 mm actually achieved, which locates the residual outside the estimator and is consistent with the detector comparison of Table 1.

Free-flight compensation

The identified offsets were applied as a feed-forward moment trim at take-off. The values flown are compared in Table 3 against what the final pipeline returns, since the pipeline was refined after the flights were performed.

The two agree to 8 mN m in the median and 34 mN m at worst, and nine of the ten components fall inside the identification scatter that was present at the time of flight; the worst is 1.3 standard deviations. Reflying would therefore command the same moments to within the precision of the identification itself, and the free-flight results are reported against the pipeline as described.

[**TODO:** Report the flight performance here: attitude tracking error and take-off transient with and without compensation, three repetitions per configuration per condition, with the paired statistic across the five configurations. This file supplies the commanded offsets only.]

Table 3: Moment offsets commanded in the free-flight tests against the final pipeline. “std” is the scatter of the identification at the time of flight.

case	comp.	flown [N m]	std [N m]	final [N m]	diff /std
case_01	$M_{\text{off},x}$	-0.064	0.037	-0.048	0.43
case_01	$M_{\text{off},y}$	0.331	0.016	0.336	0.33
case_02	$M_{\text{off},x}$	-0.485	0.009	-0.479	0.71
case_02	$M_{\text{off},y}$	0.333	0.010	0.334	0.10
case_03	$M_{\text{off},x}$	-0.282	0.028	-0.272	0.37
case_03	$M_{\text{off},y}$	-0.006	0.019	-0.023	0.89
case_04	$M_{\text{off},x}$	0.173	0.031	0.165	0.24
case_04	$M_{\text{off},y}$	-0.012	0.024	-0.007	0.21
case_05	$M_{\text{off},x}$	0.307	0.026	0.341	1.30
case_05	$M_{\text{off},y}$	0.343	0.031	0.334	0.30

Summary

The method identifies a centre-of-mass offset to 1.64 mm RMS (95% CI 0.92–2.24) without a load cell, a test stand, or any flight, from ground-contact excitation alone. It repeats to 2.9% on the onset moment, better than every model-agnostic baseline tried, although the delivered offset is insensitive to that choice. Its own modelling error is bounded a priori at 0.174 mm and measured at 0.001–0.004 mm, so what limits the accuracy is the contact geometry rather than the estimator or the number of repetitions — a conclusion the variance decomposition establishes directly, and one that determines where further work should go.